2.3 Postulates and Diagrams

Name _____ Period ____ Date _____

Geometry 2.3 · Student sheet

I can…

- identify and use postulates about points, lines, and planes.
- sketch a diagram from a description.
- tell what I can and cannot assume from a diagram.

Vocabulary · fill in the missing word

Postulate — a rule that is _____ without proof

Line perpendicular to a plane — a line that meets a plane in a point and is perpendicular to every line in the plane _____ that point

Part 1 · Discovery: why doesn't a stool wobble?

Test with pencils and paper.

Question	What we found
How many lines go through two points?	
How many planes go through three points that are not on a line?	
Why can a four-legged table wobble but a three-legged stool can't?	

Part 2 · Notes and examples (fill in each note, then work the example)

Two Point Postulate: through any two points there is exactly _____ line. **Plane Intersection Postulate:** if two planes intersect, their intersection is a _____.

Example 1

Which postulate does each statement illustrate?

a. Points A and B lie on exactly one line.

b. Planes P and Q meet in line ℓ.

Three Point Postulate: through any three _____ points there is exactly one plane.

Example 2

Explain why a three-legged stool never wobbles, using a postulate.

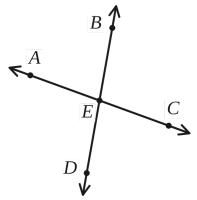
Example 3

Sketch: line ℓ is perpendicular to plane P at point A, and ray AB lies in plane P.

Reading diagrams: you CAN assume points are collinear and angles are adjacent or vertical as drawn. You CANNOT assume segments or angles are _____ or that lines are perpendicular unless it is marked.

Example 4

Use the diagram. Which statements can you assume?



a. A, E, and C are collinear.

b. ∠AEB and ∠BEC are adjacent.

c. ∠AEB is a right angle.

d. AE = EC

Example 5

A student says, "$\overline{AE} \cong \overline{EC}$ because they look the same length in the diagram." What is wrong?

Part 3 · Partner practice

Partner 1

Which postulate? "Through points P, Q, and R, which are not on one line, there is exactly one plane."

Partner 2

Two lines cross at point E in a diagram with no marks. Can you assume the angles across from each other are vertical angles? Can you assume they are right angles?